

Week 5 ADS

L1 Clustering : Recap

falls in unsupervised learning paradigm

- Input: $\{x_1, x_2, \dots, x_n\}$ $x_i \in \mathbb{R}^d$

Output: $\{z_1, z_2, \dots, z_k\}$ $z_i \in \{1, \dots, k\}$

one such way is K-Means / Lloyd's Algorithm

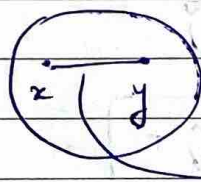
→ centroid computation

→ clusters re-assignment

→ at one pt., it should be converge.

- Property of clusters emerging from KMeans
→ clusters are in convex / conex regions.

if this is
not the case,



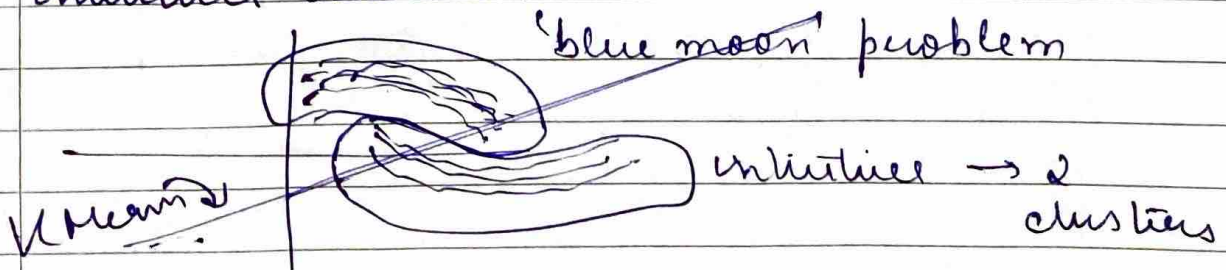
then that means

KMeans algo. has

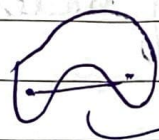
not converged accurately.

entire line segment
lies within the
region.

- Egs. where KMeans could fail in producing
intuitive clusters.

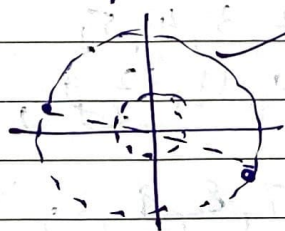


clearly, KMeans won't be able to do so, the crescent regions won't be created.



non-convex problems comes in here.
Here, KMeans fails.

- Another ex. could be of concentric circles.



the outer region is not convex.

⇓ solnⁿ

graph-based clustering (Spectral clustering)

L2

Spectral Clustering: Graph Cuts

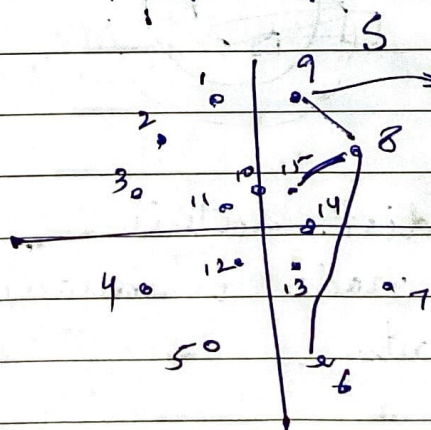
- we will treat datapts. as the nodes of the graph.

$G(V, E)$

vertex set

edges set

we will dealing with undirected edges.



consider them as vertices (nodes)

we will assign some wts. to the edges based on the distances.

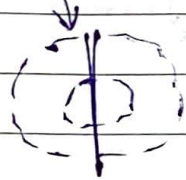
try to think of clustering as cutting a graph

in 2 pieces (for this ex.)

A cut is just a subset of nodes in the graph.

$$\text{cut}(A, S/A) = \sum_{\substack{i \in A \\ j \in B}} w_{ij}$$

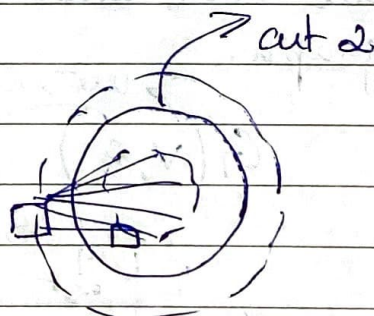
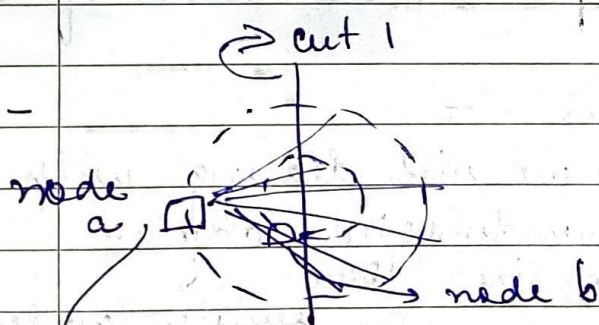
cut_{kmeans} = $\left(\begin{array}{l} \{1, 2, 3, 4, 5, \\ 10, 11, 12\} \\ \{6, 7, 8, 9, \\ 13, 14, 15\} \end{array} \right)$



$w_{ij} \rightarrow$ weight of edge $\{i, j\}$

$$w_{ij} = e^{-\frac{\|x_i - x_j\|^2}{2\sigma^2}} \rightarrow \text{HPT}$$

$w \propto \frac{1}{\text{distance}}$



as per each cut, a dist. metric.
distance for that node is measured
against the opponents.

contribⁿ of node a to cut 1 < cut 2
as dist $\uparrow$, cuts $\downarrow$

contribution of node b to cut 1 > cut 2

L3 Spectral Clustering: Min Cut

- We want cuts such that cut weights are as small as possible.

$\sum$ of distances are large $\Rightarrow$ cuts are small.

$\min_{\text{ACS}} \text{cut}(A, S|A)$ } we can think of this as assigning 0 to 1 piece and 1 to other piece.

$$\min_{f \in \{0,1\}^n} \sum_{i,j} f_i (1 - f_j) w_{ij} \quad \xrightarrow{\times} \quad \frac{-\|x_i - x_j\|^2}{2\sigma^2} > 0$$

$$= \sum_{i,j} f_i w_{ij} - \sum_{i,j} f_i f_j w_{ij}$$

$$\begin{cases} = 0 & \text{if } f = (0, 0, \dots, 0) \\ = 0 & \text{if } f = (1, 1, \dots, 1) \end{cases}$$

these cases must be avoided.

$$= \sum_{i,j} f_i w_{ij} - f^T w f$$

$$= \sum_i f_i \left(\sum_j w_{ij} \right) - f^T w f$$

$$= f^T D f - f^T w f$$

$$= f^T \underbrace{(D - w)}_L f$$

$$D = \begin{bmatrix} \sum_{j \neq 1} w_{1j} & 0 & \dots \\ 0 & \sum_{j \neq 2} w_{2j} & \\ \vdots & \vdots & \ddots \end{bmatrix}$$

$$\min_{f \in \{0,1\}^n} f^T L f$$

where $L = D - W$

- make sure $\|f\|^2 = 1 \Rightarrow$ avoids all 0s.

this is
a
NP-
hard
problem

$f^T 1 = 0 \Rightarrow$ avoids all 1s.

$$\min_{f \in \{0,1\}^n} f^T L f \quad L = D - W$$

- Let's relax our cut objective -

we have to solve this now.

$$\boxed{\begin{array}{l} \min_{f \in \mathbb{R}^n} f^T L f \\ \|f\|^2 = 1 \\ f^T 1 = 0 \end{array}} \xrightarrow{\text{solve}}, f^* \in \mathbb{R}^n$$

we won't have the perfect mapping as by $f \in \{0,1\}^n$. At least they will be mapped to noisy versions of 0 & 1. Then just run the kMeans on f^* .